WORKED SOLUTION

STEP I 2005, Question 2

Pure • Parabola Tangents and Normals

STEP I 2005, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q2

The key idea

Parametrising the parabola $y^2 = 4x$ by $(t^2, 2t)$ makes every tangent and normal a short linear equation in x and y, so finding intersections is just solving two simultaneous linear equations. The condition that $(1, 0)$, the focus, lies on PQ turns into a single neat relation between p and q, and that relation is exactly what makes the tangents at P and Q perpendicular. Once you have that, the rectangle almost draws itself.

Solution

The tangents and the point R

At the point $(t^2, 2t)$ we have $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$ and $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 2$, so the gradient of C there is $\dfrac{2}{2t} = \dfrac{1}{t}$ (fine, since $t \neq 0$). The tangent at P is therefore

$$y - 2p = \frac{1}{p}(x - p^2) \quad \Longleftrightarrow \quad py = x + p^2,$$

and likewise the tangent at Q is $qy = x + q^2$. Subtracting,

$$(p - q)\, y = p^2 - q^2 = (p - q)(p + q),$$

and since $p \neq q$ we may divide by $p - q$ to get $y = p + q$. Then $x = py - p^2 = p(p + q) - p^2 = pq$. So

$$R = (pq,\ p + q),$$

as required.

The normals and the point S

The normal at $(t^2, 2t)$ is perpendicular to the tangent, so it has gradient $-t$. The normal at P is

$$y - 2p = -p\,(x - p^2) \quad \Longleftrightarrow \quad px + y = 2p + p^3,$$

and the normal at Q is $qx + y = 2q + q^3$. Subtracting,

$$(p-q)\,x = 2(p-q) + (p^3 - q^3) = (p-q)(2 + p^2 + pq + q^2),$$

so, dividing by $p - q \neq 0$ again,

$$x = p^2 + pq + q^2 + 2, \qquad y = 2p + p^3 - px = -p^2q - pq^2 = -pq\,(p+q).$$

These hold for any p and q; we now use the extra condition.

Using the condition that $(1,0)$ lies on PQ

The chord PQ has gradient

$$\frac{2p - 2q}{p^2 - q^2} = \frac{2}{p+q}$$

(when $p + q \neq 0$), so its equation is $y - 2p = \dfrac{2}{p+q}(x - p^2)$, which simplifies to

$$(p+q)\,y = 2x + 2pq.$$

This form also covers $p + q = 0$, when it gives the vertical line $x = -pq = p^2$ through P and Q. Putting $(x, y) = (1, 0)$ gives $0 = 2 + 2pq$, so

$$pq = -1.$$

Substituting into the coordinates of S:

$$S = (p^2 + q^2 + 2 + pq,\ -pq(p+q)) = (p^2 + q^2 + 1,\ p+q),$$

as required. Notice also that now $R = (-1,\ p+q)$, which lies on the line $x = -1$, the directrix.

$PSQR$ is a rectangle

> **Ask yourself** What is the cleanest way to prove that four given points form a rectangle?

A good route is: show it is a parallelogram (the diagonals bisect each other), then show one angle is a right angle.

Parallelogram. In $PSQR$ the diagonals are PQ and SR. Using $pq = -1$,

$$R + S = (pq + p^2 + q^2 + 1,\ 2(p+q)) = (p^2 + q^2,\ 2p + 2q) = P + Q,$$

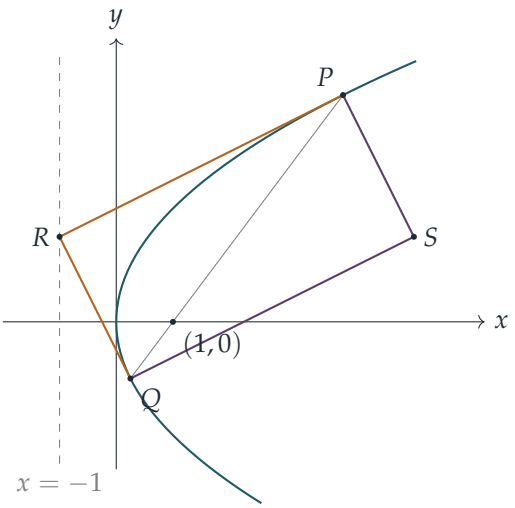
so the midpoint of RS equals the midpoint of PQ. The diagonals bisect each other, so $PSQR$ is a parallelogram. It is a genuine one, not a degenerate one: P, R and Q are not collinear, because R lies on both tangents and the tangents at two different points of a parabola meet only once and neither passes through the other point of contact.

A right angle. The sides RP and RQ lie along the tangents at P and Q, with gradients $\frac{1}{p}$ and $\frac{1}{q}$. Their product is

$$\frac{1}{p} \cdot \frac{1}{q} = \frac{1}{pq} = -1,$$

so the two tangents are perpendicular and the angle PRQ is a right angle.

A parallelogram with one right angle is a rectangle, so $PSQR$ is a rectangle. (Equivalently: the angles at P and Q are right angles because each normal is perpendicular to its own tangent, and with the right angle at R the fourth angle must be one too.)



The figure shows the case $p = 2$, $q = -\frac{1}{2}$: tangents in amber, normals in plum, and the focal chord PQ in grey.

Checks

Take $p = 2$, $q = -\frac{1}{2}$, so $pq = -1$. Then $P = (4, 4)$ and $Q = (\frac{1}{4}, -1)$, and the line through them, $y - 4 = \frac{4}{3}(x - 4)$, does pass through $(1, 0)$. The formulae give $R = (-1, \frac{3}{2})$ and $S = (\frac{21}{4}, \frac{3}{2})$. Check directly: $\overrightarrow{RP} = (5, \frac{5}{2})$ and $\overrightarrow{QS} = (5, \frac{5}{2})$ are equal, and $\overrightarrow{RQ} = (\frac{5}{4}, -\frac{5}{2})$ has dot product $\frac{25}{4} - \frac{25}{4} = 0$ with $\overrightarrow{RP}$. So $PSQR$ is indeed a rectangle. We also checked all the coordinate formulae with exact fractions for about two thousand random choices of p and q.

Common slips

- Dividing by $p - q$ without saying why it is allowed; the question gives $p \neq q$ precisely so you can.
- Showing only that adjacent sides are perpendicular at one corner, which does not

by itself make a rectangle: you need either a parallelogram plus one right angle, or right angles at three corners.

- Forgetting to use $pq = -1$ in S: the general normal intersection is $\big(p^2 + pq + q^2 + 2,\ -pq(p+q)\big)$, and it only simplifies to the given form on the focal chord.

Beyond STEP

A chord through the focus is a *focal chord*, and this question proves two classical facts about it: the tangents at its ends are perpendicular and meet on the directrix. Both are special cases of the theory of poles and polars for conics, in which the directrix is the polar of the focus; projective geometry makes such results almost automatic. A good place to explore parametric methods for conics in this spirit is Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.